

DukaanPayAI

“Vyapaar ka Naya Vishwas”

Autonomous AI Business multi-agentic system for 30 Million+ Indian Kiranas

Transforming passive Soundboxes into an autonomous intelligence system—anticipating stockouts, predicting customer churn, and orchestrating revenue growth entirely over [WhatsApp](#) and [Indic Voice](#) with zero new apps or manual data entry.

PRESENTED BY

Team Sentinels (Akshat Arya & Mayank Yadav)

PROJECT NAME

AI Business Partner for Merchants

IDEA / PRODUCT NAME

DukaanPayAI

THEME / TRACK SELECTION

Merchant Growth AI (Track 01)

The Problem & The Kirana Economy

30 Million+ Kirana stores power India's retail, yet remain trapped in blind spots and passive payment traps.



Rajesh Gupta, Store Owner

Gupta Kirana Store • Kanpur, UP

- ⚠️ ₹4,200 lost weekly to avoidable evening stockouts on top 5 FMCG SKUs.
- ⚠️ 40% of regular buyers churn to quick-commerce dark stores without notice.
- ⚠️ ₹14-₹25/unit overcharges by distributors undetected on paper invoices.

"14-hour days behind the counter with zero digital assistant or real-time guidance."

The 3 Structural Failures

01 / PASSIVE TRAP

Soundbox Only Confirms Payments

Payment audio boxes confirm incoming transactions, but offer zero proactive intelligence, inventory alerts, or customer retention loops.

02 / POS SOFTWARE FRICTION

92% Abandon Complex ERPs

Heavy desktop POS and mobile inventory apps demand manual barcode scanning and tedious data entry that merchants abandon within 14 days.

03 / NO ACTION ENGINE

Analytics Without Execution

SaaS dashboards display passive analytics charts. Overworked store owners need 1-tap autonomous execution directly where they communicate.

The ₹250B+ Opportunity

30M+

Kirana Stores

₹85B+

Daily UPI Vol

92%

Zero Tech

TAM (TOTAL MARKET)

₹250B+ (\$3.0B)

30M+ Indian MSMEs seeking digitized growth, smart inventory, and autonomous sales assistance.

SAM (SERVICABLE)

₹85B (\$1.0B)

10.8M active Paytm Soundbox and QR merchants with high-frequency daily transaction streams.

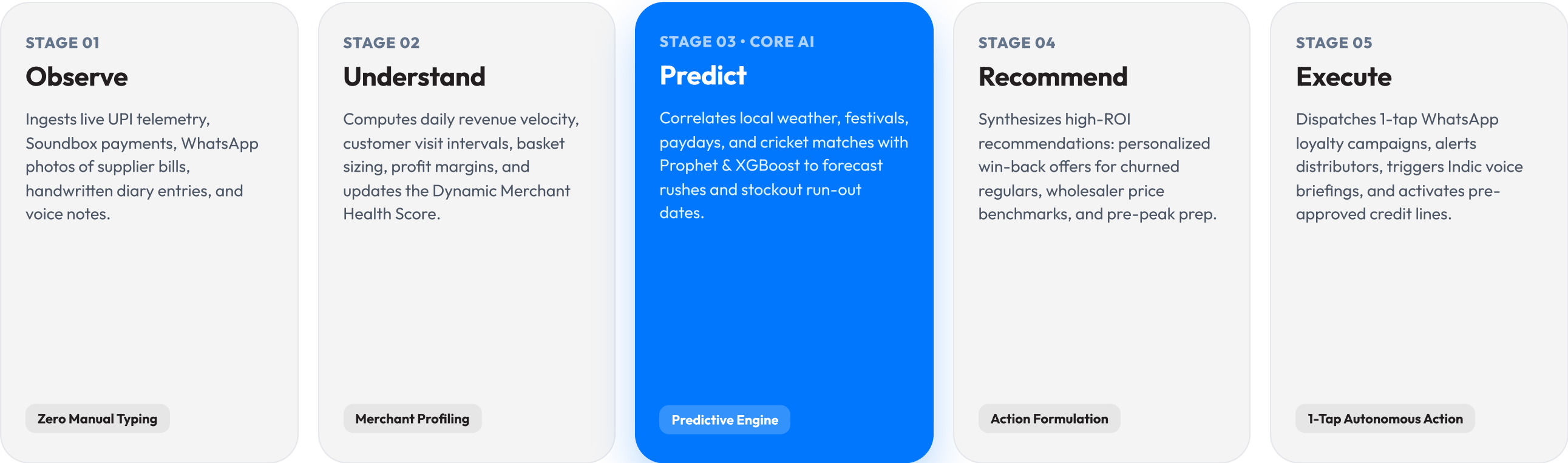
SOM (TARGET Y3)

₹32B (\$380M)

4.2M high-velocity tier 1-3 Kiranas, grocery, and pharmacy retailers onboarded within 36 months.

The Autonomous Merchant Growth Loop

Paytm evolves from a passive payment processor into an autonomous AI business partner that drives revenue.



 **Zero App Fatigue Philosophy:** The merchant never installs a new app or navigates confusing web dashboards. Every notification, alert, and 1-tap approval happens natively inside **WhatsApp** and through **Indic AI Voice Calls**.

100% NATIVE ADOPTION

What We Provide & Kirana User Flow

A Day in the Life: How the Paytm Growth Agent Partners with Store Owners from 06:30 AM to 10:30 PM

FEATURE ECOSYSTEM

Core Capabilities Provided

- ✓ **Proactive WhatsApp Intelligence:** Automated daily morning & evening briefings delivered into existing habits—zero new apps.
- ✓ **Real-Time UPI Telemetry:** Live stream monitoring payment velocity, customer purchase frequency, and sudden volume anomalies.
- ✓ **Pre-Rush Footfall Warning:** Anticipates peak rush hours 2 hours in advance to prep fast-moving items and avoid stockouts.
- ✓ **Automated Retention Campaigns:** AI identifies churn risks and sends 1-tap WhatsApp cashback offers driving repeat visits.
- ✓ **Multimodal Shelf & Bill OCR:** Snaps of shelves or supplier invoices automatically update stock and draft best-rate wholesaler POs.

 **Emergency & Business Safety Net:** In unexpected sales crashes, supplier disputes, or working capital crunches, the AI activates emergency protocol—instant pre-approved Paytm credit line + alternative wholesaler dispatch.

REAL-WORLD SCENARIO

A Day with Rameshji (Gupta Kirana Store)

06:30 AM

Wakeup Briefing

Rameshji checks WhatsApp: **₹18,450 sales yesterday** (+18%). 42°C heatwave alert predicts **+35% cold drink spike**; advises stocking ice & 3 crates of cola before opening.

11:00 AM

Real-Time Telemetry

AI tracks 85+ live Paytm QR payments. Ingests payment velocity silently; flags an anomalous ₹14 wholesaler overcharge on cooking oil benchmarked across 5 local stores.

04:30 PM

Pre-Peak Rush Alert

Proactive alert 2.5 hours before 7 PM rush: *“Evening milk & bread stock low. Tuesday footfall surges in 150 mins—restock front counter now to prevent ₹2,800 stockout loss.”*

06:00 PM

1-Tap Retention Offer

AI detects 18 weekly regulars slipping away (12 days absent). Generates personalized 5% cashback message. Rameshji reviews on WhatsApp and taps **“Approve & Send”** in 1 second.

08:30 PM

Shelf OCR & Reorder

Rameshji snaps a photo of empty Atta shelves. Vision OCR extracts deficit (40 kg), benchmarks cheapest distributor (₹38/kg), and prepares purchase order for 1-tap reorder.

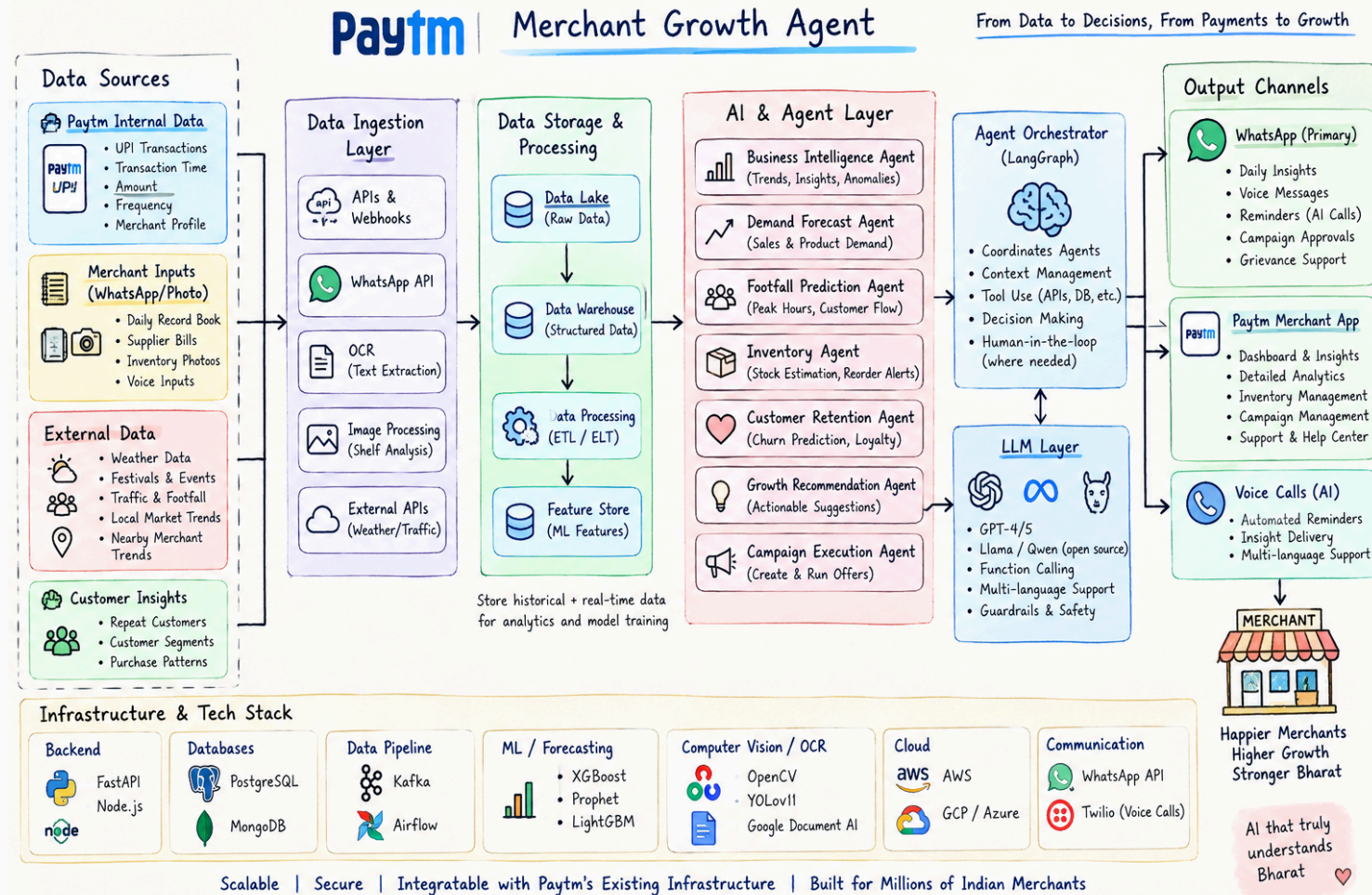
10:30 PM

Night Closing & Safety

Day closing P&L: ₹24,200 revenue. Weekly Growth Health Score rises to 88/100. Emergency credit safety net active with instant ₹2,50,000 Paytm business loan ready on tap.

End-to-End System Architecture

LangGraph Multi-Agent Orchestration & Edge Micro-Services Architecture



Enterprise multi-agent pipeline: Ingests Paytm UPI telemetry, WhatsApp multimodal OCR, predictive XGBoost/Prophet models, and LangGraph LLM supervisor into proactive loops across WhatsApp, Merchant App, and AI Voice Calls.

Multi-Agent Intelligence & Enterprise Stack

Specialized autonomous agents coordinated by LangGraph, built on resilient Bharat-scale cloud infrastructure.



Business Intelligence Agent

Monitors revenue velocity, flags payment volume dips, computes gross margins, and identifies high-value customer spending trends.



Inventory & Supplier Agent

Predicts stockouts 3-5 days ahead via Prophet time-series models, audits supplier invoices with OCR, and benchmarks regional prices.



Voice & Telephony Agent

Conducts low-latency interactive phone calls in Hindi, Marathi, Tamil, Bengali, and Hinglish via Bhashini & Twilio edge connectors.



Customer Retention Agent

Tracks buyer purchase intervals, identifies churn risks before they leave, and crafts targeted WhatsApp cashback loyalty incentives.



Campaign Execution Agent

Auto-generates high-converting WhatsApp cashback templates and executes marketing blasts immediately upon 1-tap merchant approval.



LangGraph Orchestrator

Supervisor agent managing state memory, context routing, tool calling, validation checks, and human-in-the-loop safety verification.

• Sub-500ms Edge Latency

• Kafka & Redis (50k+ TPS)

• PostgreSQL + pgvector Embeddings

• Llama-3-Vision & YOLOv11 OCR

• RBI Sovereign Cloud Compliance

Multimodal Perception & Indic Voice AI

Zero-friction data capture: Voice notes, handwritten diaries, shelf photos, and natural Indic phone conversations.



Paper Supplier Bills

Vision OCR parses blurry carbon-copy wholesaler receipts and printed bills snapped on WhatsApp in under 3 seconds.

Extracted: Fortune Oil (15L)
Rate: ₹142/L vs City Avg ₹128/L
⚠ Alert: ₹14/L Overcharge
Flagged



Handwritten 'Khaata'

Interprets cursive handwritten customer credit ledgers in Hindi, Gujarati, and local scripts with spatial layout parsing.

Parsed: Sharma Ji Udhaar
Amount: ₹1,850 (Due: 18 days)
Action: Gentle WhatsApp reminder
drafted



Shelf Vision Scans

Merchant snaps a quick photo of store shelves or refrigerators. AI detects empty spaces and estimates stockout dates.

Object Detection: Atta Shelf
Stock Level: 8kg remaining
⚡ Forecast: Stockout in 28 hrs



10+ Indic AI Voice Calls

Interactive telephony in Hindi, Marathi, Tamil, Telugu, and Hinglish for audio briefings and voice-to-action confirmations.

Audio Call: "Namaste Rameshji, kal sham ka rush 30% badhega. Kya hum 3 peti cola reorder karein?"
Merchant: "Haan, kar do." (Done)

WhatsApp as the Merchant OS & Predictive Engine

Where intelligence turns into revenue: 1-tap conversational execution coupled with multi-signal demand forecasting.

CONVERSATIONAL ACTION INTERFACE

01 • MORNING BRIEFING

07:30 AM

"Good morning Rameshji! Yesterday's revenue was **₹18,450 (+18%)**. Heatwave alert today: cold beverage demand will surge **+35%**. Restock 3 crates of cola before 11 AM."

02 • STOCKOUT ALERT

02:15 PM

" **Stockout Warning:** Basmati Rice runs out in 36 hours. Best distributor rate benchmarked at **₹38/kg**. Tap to place purchase order."

Approve Reorder (1 Tap)

03 • WIN-BACK OFFER

05:45 PM

"18 regular customers haven't visited in 12 days. Send them a personalized **5% cashback offer** on purchases over ₹200 to win them back?"

Approve & Send (1 Tap)

Multi-Signal Demand Predictor

Prophet & XGBoost models synthesize 5 live external data signals into continuous 7-day hourly demand forecasts:

 **Weather:** Heatwaves (+35% drinks), Monsoon (+50% tea/biscuits)

 **Festivals:** Diwali (+60% dry fruits/sweets), Holi (+40% snacks)

 **Live Events:** IPL cricket matches (+45% chips/cold drinks)

 **Payday Cycles:** 1st–7th of month (+28% high-margin staples)

Lost Revenue Leakage Detector (₹12,450/mo Recovered)

₹4,200

Stockout Recovery

Preventing lost sales on high-turnover milk, bread, and beverage surges.

₹6,800

Churn Recovery

Winning back lapsed regulars with automated 1-tap WhatsApp cashback.

₹1,450

Supplier Price Audit

Benchmarking distributor bills to block invisible wholesale overcharges.

Merchant Health Score & Paytm Lending Moat

Transforming payment telemetry into proprietary credit underwriting and collateral-free merchant financing.

DYNAMIC CREDIT UNDERWRITING

Merchant Health Score



STATUS: EXCELLENT TIER

PRE-APPROVED CREDIT LINE

₹2,50,000

Repayment: ₹350/day via daily UPI settlement escrow.
100% paperless disbursement in 90 seconds.

35%

Payment Velocity & Stability

Evaluates 30-day rolling UPI volume, ticket size dispersion, daily cashflow predictability, and peak hour velocity.

25%

Inventory Turnover & Health

Monitors stockout recurrence, replenishment consistency, multimodal invoice audits, and shelf availability ratios.

20%

Customer Retention & Loyalty

Tracks repeat customer ratios, churn recovery speed via 1-tap WhatsApp campaigns, and Customer Lifetime Value (CLV).

20%

Supplier Settlement Discipline

Analyzes invoice payment clearance promptness, dispute frequency, and wholesaler price audit verification integrity.

 **Paytm's Unfair Lending Moat:** Traditional banks reject Kiranas due to lack of balance sheets. Paytm possesses live ground-truth cashflow telemetry and multimodal inventory health. This enables pre-underwritten credit lines with **40% lower default rates** and automated escrow collection on daily settlements with zero CAC.

THE VISION AHEAD

The Future of Indian Retail: Autonomous Kiranas

Empowering 30 Million+ Indian merchants with enterprise AI—without requiring new hardware, apps, or behavioral changes.

3-Year Monetization (ARR Growth)



- **SaaS Subscription:** ₹299–₹999/mo tiered merchant intelligence tier.
- **Campaign Take-Rate:** 1.5%–2% on incremental revenue driven via WhatsApp.
- **Lending Origination:** 1.5%–2.5% origination fee on underwritten credit lines.

18-Month Scaling Roadmap

PHASE 1: MVP

UPI telemetry, daily briefings, anomaly alerts.

PHASE 2: OCR

Paper bills, khaata diaries, shelf vision.

PHASE 3: AGENTS

LangGraph multi-agent demand forecasting.

PHASE 4: VOICE

1-tap campaigns & 10+ Indic voice calls.

PHASE 5: SCALE

Bharat-wide rollout & lending credit lines.

Presented by

Team Sentinels (Akshat Arya & Mayank Yadav)

Hackathon Track

Merchant Growth AI • Paytm Build for India

Repository & Contact

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